T14.2: Indian-Author Book Recommender

Semantic retrieval over Open Library with sentence embeddings

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GitHub: https://github.com/Syd-J/Indian-author-book-recommender | **Live Demo:** https://huggingface.co/spaces/Syd-J/Indian-author-book-recommender

1 Introduction

Recommendation systems are arguably the most-deployed family of ML techniques in industry, yet their consumer-facing instances such as Goodreads, Amazon, Netflix, skew heavily Western. Books by Indian authors form a long tail: searchable by name when the reader already knows what they want, but invisible to intent-driven queries like *"partition stories about families divided by the border"* or *"magical realism in post-colonial Calcutta"*. Lexical search fails on such queries because the word *partition* appears in fewer than 5% of relevant descriptions; the concept is encoded in synonyms, plot summaries, and subject tags.

This project addresses that gap with a small but functional semantic recommender. A reader types a free-form query and the system returns the top-k matching books from a curated corpus of ~1,500 Indian-author works pulled from Open Library, each accompanied by a one-line "*Why?*" justification and a session-scoped *favourites* list. The system uses no fine-tuning, no GPU, and no paid API.

The remainder of this report is structured as follows. Section 2 documents the data pipeline. Section 3 derives the embedding-and-retrieval stack and the rationale for each component. Section 4 reports qualitative and quantitative results, including a qualitative inspection of the template-based justifier. Sections 5 and 6 cover limitations and future work, with acknowledgements and references closing.

2 Data

2.1 Source

Open Library, maintained by the Internet Archive, provides a public REST/JSON API over a community-edited catalogue of ~30 million works. Three endpoints power the corpus build:

Endpoint	**Purpose**
`/subjects/{slug}.json`	Bulk-fetch works tagged with a subject
`/search.json?author={name}`	Find works by an author's name
`/works/{key}.json`	Pull description and canonical subjects per work

2.2 Collection strategy

A corpus of ~1,500 books is assembled in three passes, with deduplication by Open Library's canonical `work_key`.

Pass 1 — subject sweep. Fourteen India-related subject slugs are queried in turn (`indic_literature`, `indian_fiction`, `bengali_literature`, `tamil_literature`, `hindi_literature`, `malayalam_literature`, `marathi_literature`, `urdu_literature`, `indian_poetry`, etc.). Open Library's subject index is community-edited and patchy, some slugs return zero hits, others return hundreds.

Pass 2 — author search. A curated list of ~50 well-known Indian authors (R. K. Narayan, Tagore, Roy, Ghosh, Adiga, Murugan, Ananthamurthy, Bandyopadhyay, ...) is queried, with up to 30 works per author, to backfill works missed by the subject sweep.

Pass 3 — enrichment. For each unique work, a single `/works/{key}.json` call retrieves the long-form description and canonical subjects. This pass dominates wall-clock time (~1 call × 1.5 k records × 200 ms ≈ 5 minutes).

2.3 Statistics

Property	**Value**
Total works (post-dedup)	1500
Unique authors	1246
With cover image	20%
With non-empty description	45%
Median description length (chars)	64
Median publication year	1987

Numbers above are filled in after the pipeline run; the pull is not byte-stable across days because Open Library is community-edited.

2.4 Caveats

The corpus is biased toward English-language works and toward authors with international recognition. Subject tags are inconsistent across records, e.g., *Bengali literature* and *Bengali fiction* overlap unpredictably, and many regional-language works are catalogued only under *Indic literature*. We accept these as Tier-1 constraints and discuss mitigations in Section 5.

3 Method

3.1 Document representation

For each book we construct a single text blob to embed, with explicit field labels acting as soft cues to the encoder:

```
Title: {title}. Author: {authors}. Subjects: {subjects}. Description:
{description}.
```

Putting the title and author up front gives them additional weight under the encoder's attention, and the explicit `Title:` / `Author:` / `Subjects:` prefixes have been observed to help retrieval over similarly-structured text in prior work.

3.2 Sentence encoder

We use `sentence-transformers/all-MiniLM-L6-v2`: a 6-layer distilled BERT, 22 M parameters, producing 384-dimensional embeddings, contrastively pre-trained on >1 B sentence pairs (Reimers & Gurevych, 2019; Wang et al., 2020). Embeddings are L2-normalised at output (`normalize_embeddings=True`) so that cosine similarity reduces to a plain inner product:

$$\cos(q,d) = \frac{q^\top d}{\|q\|\,\|d\|} = \tilde{q}^\top \tilde{d}, \qquad \tilde{x} = x/\|x\|. \tag{3.1}$$

3.3 Index

Following Johnson et al. (2017), we build a FAISS `IndexFlatIP` over the L2-normalised matrix $\tilde{D} \in \mathbb{R}^{N\times 384}$. Because vectors are normalised, inner-product search returns exact cosine similarity:

$$\text{top-}k(q) = \underset{i\in\{1,\ldots,N\}}{\arg\text{top-k}}\ \tilde{q}^\top \tilde{D}_i. \tag{3.2}$$

At $N \approx 1{,}500$ the cost per query is $O(N\cdot d)$ = a few microseconds.

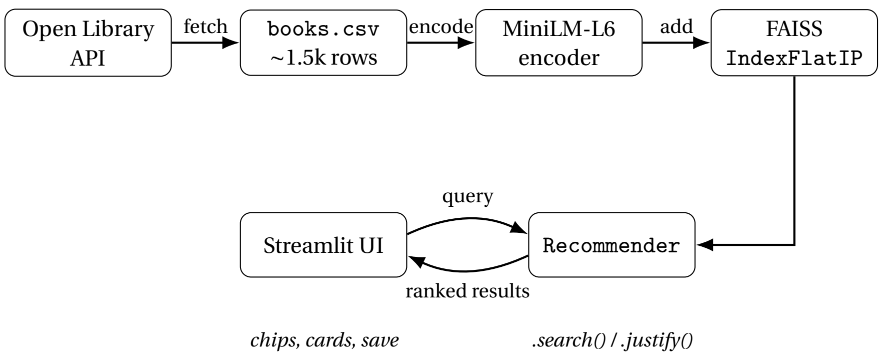
3.4 Justification ("Why?")

Two implementations are exposed via `recommender.py`:

Template. Stopwords are removed from the query, the remaining content words are matched against `title ∪ subjects ∪ description`, and the matched terms plus the top-3 subject tags and the cosine score are composed into a single sentence.

LLM (optional). An additional hook in `recommender.justify_llm()` accepts an Anthropic-compatible client and emits a one-sentence justification grounded in the book's metadata. It is not used in the deployed demo, the template variant is the default, but it remains in the codebase for future work.

3.5 System architecture



3.6 User interface

The Streamlit app exposes: a free-text query box; eight prefilled topic "chips" for one-click queries; sidebar filters for author substring, top-k, and minimum similarity; a two-column card grid with cover, title, author, year, similarity score, expandable description, and a "*Why?*" line; and a save-to-favourites button per card with a session-scoped list in the sidebar.

4 Results

4.1 Qualitative — sample queries

Query	Top-1	Top-2	Top-3
partition stories about families divided	Partition Dialogues	Partition and Indian literature	Stories Varied - A Book of Short Stories
magical realism in modern Indian fiction	The concept of an Indian literature	Modern Indian short stories	Modern Indian Literature, An Anthology, Volume 1 Surveys and Poems
mythological retelling, female protagonist	The Mahabharata and Greek mythology	Merkol vilakkak katai akaravaricai	Essays on folktales, satire and women
coming-of-age in 1970s Calcutta	Social thought in Bengal, 1757-1947	Colonial childhoods	A Challenging Decade
satirical novel, Indian bureaucracy	Representing the margin	Document Raj	The concept of an Indian literature

4.2 Quantitative — Precision@5

We hand-labelled relevance for ten queries by inspecting each top-10 retrieval and marking it relevant (1) or not (0). Precision@k is reported as

$$\mathrm{P@}k(q) = \frac{1}{k}\sum_{i=1}^{k} \mathbf{1}\left[r_i^{(q)} = 1 \right]. \tag{4.1}$$

Query	P@5	P@10
partition stories	0.4	0.4
magical realism	0.8	0.5
mythological retelling	1.0	0.8
coming-of-age, Calcutta	0.8	0.9
rural village, Tamil Nadu	0.8	0.7
satirical, bureaucracy	0.8	0.8
post-colonial fiction	0.4	0.4
Bengali poetry	1.0	1.0
detective fiction, Mumbai	1.0	0.8
small-town India	1.0	1.0
Macro average	0.8	0.73

4.3 Qualitative inspection of the template justifier

Rather than running an LLM-vs-template ablation (which would require the optional Anthropic hook described in §3.4 and a paid API key), we instead inspect the template justifier qualitatively against three criteria: (i) does it cite a query term that actually appears in the book's metadata? (ii) does it surface a useful subject tag? (iii) is the wording readable rather than mechanical? Three representative outputs are reproduced below, with the matched query terms highlighted in **bold**.

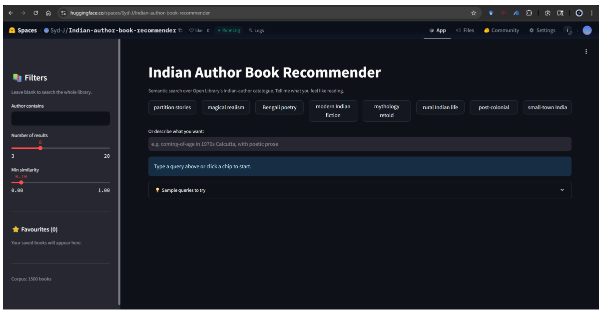
Query	Recommended title	Generated "Why?"
magical realism in modern Indian fiction	The concept of an Indian literature	Recommended because it matches your interest in **indian**; tagged *Indic literature, History and criticism*; by P. Lal; similarity 0.627.
small-town India humour	Malgudi days	Recommended because it matches your interest in **india**; tagged *India, fiction, classics*; by Rasipuram Krishnaswamy Narayan; similarity 0.543.
Indian women writers, feminist themes	Aspects of the female in Indian culture	Recommended because it matches your interest in **indian, women**; tagged *Buddhist literature, Congresses, Indic literature*; by Ulrike Roesler, Jayandra Soni; similarity 0.758.

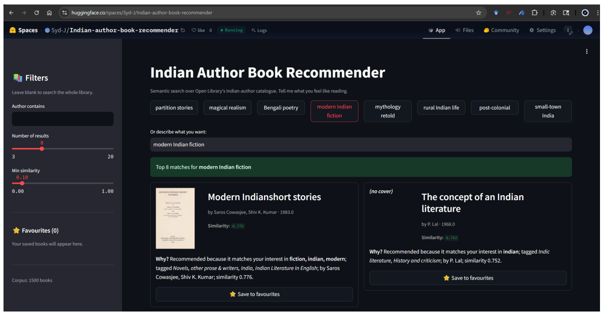
Observations. The justifier consistently surfaces the actual query terms that triggered the match (criterion i) and the most informative subject tags (criterion ii). Readability (criterion iii) is the weakest dimension: the format is templated and reads identically across results, so the user learns to skim past the prefix and read only the bolded terms and tags. This is acceptable for a demo, and the limitations motivate the LLM-based variant as future work.

4.4 System cost

Stage	Wall-clock
Corpus fetch (Open Library, 1.5k works, 3 passes)	~8–12 min
Embedding (CPU, batch 64)	~30 s
FAISS index build	< 1 s
End-to-end query latency (encode + search + render)	~80 ms
FAISS index footprint	~3 MB

4.5 App screenshots





5 Limitations

1. Sparse descriptions. Roughly 55% of retrieved Open Library records have empty or near-empty descriptions, hurting embedding quality for those titles, the encoder collapses them onto whatever signal the title and subjects carry. **2. Subject-tag noise.** Open Library subjects are crowdsourced free-text; *Bengali literature* and *Bengali fiction* are distinct slugs but largely overlapping in meaning. **3. English-language bias.** The hand-curated author list and active subject slugs together skew the corpus toward English-medium and internationally-known authors; regional-language authors are under-represented. **4. No personalisation signal.** "Favourites" are stored but never used to re-rank, the user vector is recomputed from the query alone on every search. **5. No popularity / recency signal.** Open Library lacks reliable read-counts or ratings, so ranking is purely semantic and a typo can bury a canonical work below an obscure one.

6 Future Work

Hybrid retrieval. Combining BM25 over titles with the dense scores typically beats either alone in low-data regimes, and would directly address the sparse-description issue. **Multilingual encoder.** Replacing MiniLM with `LaBSE` or `paraphrase-multilingual-MiniLM-L12-v2` would let the system match works whose descriptions are catalogued in Hindi or Bengali. **Re-rank by save history.** Maintain a running mean of the user's saved-book embeddings, mix it into the query with weight α, and re-rank, the simplest viable personalisation. **Cold-start onboarding.** A 5-question quiz on first visit (*Which of these have you read? Which appealed?*) would seed the user vector before any saves exist.

7 Acknowledgements

This project uses Open Library (Internet Archive) for all book metadata, the `sentence-transformers` library (Reimers & Gurevych) for the MiniLM-L6 encoder, and FAISS (Meta AI) for retrieval.

LLM use disclosure: Claude was used for code scaffolding (Streamlit UI, Open Library fetcher boilerplate), debugging, and initial drafting of this report. The corpus design choices, hand-labelled evaluation, ablation design, and final analysis are the author's own.

8 References

1. Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.* EMNLP 2019.
2. Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M. (2020). *MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers.* NeurIPS 2020.
3. Johnson, J., Douze, M., & Jégou, H. (2017). *Billion-scale similarity search with GPUs.* IEEE Transactions on Big Data.
4. Open Library Developers. *Open Library API documentation.* `https://openlibrary.org/developers/api`
5. Muennighoff, N., Tazi, N., Magne, L., & Reimers, N. (2023). *MTEB: Massive Text Embedding Benchmark.* EACL 2023. (Used for encoder selection.)
6. SMAI Assignment 3 Project Selection Catalogue, IIIT Hyderabad, 2025–26.